



SpiderVault 60.0

60 kVA Three-Phase Solar Hybrid UPS

SPECIFICATION SHEET

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BATTERY INPUT	
Battery Cell Technology	Lithium Ion
Battery Capacity	60 kWh
Quantity of Batteries	14 Nos.
Nominal Voltage (each battery)	48 V
Full Charging Voltage	54.3 V
Full Discharging Voltage	40 V
Max. Continuous Charging Current	150 A
Max. Continuous Discharging Current	90 A × 14 batteries
BMS	5th Gen BMS · DC Breaker

PV STRING INPUT	
Max. DC Input Power	66 kW (12 × 5.5 kW)
No. of MPPT Trackers	12
Max. PV Array Open Circuit Voltage	450 VDC
MPPT Voltage Range	90 – 450 VDC
Max. PV Input Current (per MPPT)	18 A
Max. Solar Charge Current	450 A

AC OUTPUT			
Rated AC Output Power	60 kW (60,000 W)	Grid Type	Three Phase (3P/3P)
Surge Power	60,000 VA (equal to rated)	Transfer Time	10 ms
AC Output Voltage (Battery Mode)	415 VAC ± 5%	Waveform	Pure Sine Wave
Max. Continuous AC Passthrough	200 A	Switching Modes	On-Grid / Off-Grid / Hybrid
Output Frequency	50 Hz / 60 Hz (Auto sensing)		

EFFICIENCY	
Peak Efficiency	~94%
MPPT Efficiency	≥ 99%

PROTECTION	
PV Input Lightning Protection	Yes
Anti-Islanding Protection	Yes
PV String Reverse Polarity Protection	Yes
Over Current Protection	Yes
Output Shorted Protection	Yes
Temperature Protection	Yes
Output Over Voltage Protection	DC Type II AC Type III
DC Switch	Yes

GENERAL DATA	
Dimensions W × D × H	815 × 1155 × 1375 mm
Net Weight	690 ± 5 kg
Mounting / Installation	Floor Standing
Topology	High Frequency Isolation
IP Protection	IP65
Operating Temperature	-10°C to 60°C
Noise Level	< 50 dB
Cooling	Advanced Nano PCM + Forced Convection
Display	LCD — Operating mode / Load / Input / Output
PV Connector	MC4
AC Connector	Terminal Connectors
Communication Interface	BMS CAN Bluetooth Wi-Fi AI Cloud Dry-contact
Warranty	5 Years Standard · 12 Years Extended

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CERTIFICATIONS & STANDARDS

INDIA — BIS CERTIFIED

BIS R-63005436

IS 16221 (Part 2): 2015 / IEC 62109-2: 2011

IS 16169: 2019 / IEC 62116: 2014

IS 17980: 2022

SAFETY REGULATION

IEC/EN 62109-1, IEC/EN 62109-2

GRID REGULATION

IEC 61727, G99, G98

Anti-islanding: Integrated

60 kW

Rated Power

~94%

Peak Efficiency

60 kWh

Battery Capacity · 14 units

12

MPPT Trackers